

External Design Document for Q2-Commerce

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1. Introduction

- 1.1 Purpose
- Q2-Commerce is a robust platform for managing e-commerce operations using React.js, Node.js, and Firebase Firestore.
- 1.2 Scope
 - - Architecture (Frontend & Backend)
 - - Database schema & workflows
 - - API design, security, and scalability plans

2. System Architecture

- 2.1 Overview
 - - Frontend: React.js
 - - Backend: Node.js with Express.js
 - - Database: Firebase Firestore
- 2.2 Workflow
 - - Frontend: Dynamic components
 - - Backend: Logic & DB interaction
 - - Database: Real-time sync

3. Backend Structure

- 3.1 Directory Breakdown
 - config/, controllers/, models/, routes/, services/, server.js, .env, README.md
- 3.2 Example Workflow - Cart Checkout
 - 1. Frontend POST /transactions
 - 2. Backend validates and processes
 - 3. Firestore updates records

4. Frontend Structure

- 4.1 Directory Breakdown
 - assets/, styles/, components/, context/, hooks/, pages/, services/, utils/, App.jsx, index.jsx
- 4.2 Component Examples
 - - ProductCard.jsx: Product + Add to Cart
 - - ThemeToggle.jsx: Theme switcher

5. Firestore Data Schema

- 5.1 Hierarchical Structure
 - /users/, /products/, /orders/, /messages/, /transactions/, /reviews/
- 5.2 Example Documents
 - - User: name, email, role, shopping_cart
 - - Product: name, price, vendor, stock

6. Key Features

- 6.1 User Management
 - - Role-based profiles and editing
- 6.2 Transactions
 - - Create/update transactions, adjust stock
- 6.3 Messaging System
 - - Real-time chat with read/delete

7. API Endpoints

- Users
 - POST /users → create user
 - PUT /users/:id → update user
- Transactions
 - POST /transactions → create transaction

8. Security

- 8.1 Firestore Rules
 - - Role-based read/write access
- 8.2 Encryption
 - - Passwords & sensitive data encrypted

9. Scalability

- 9.1 Firestore Optimization
 - - Indexing, sharding for high performance
- 9.2 Modular Architecture
 - - Easy feature addition & maintenance

10. Future Enhancements

- 10.1 AI-Powered Recommendations
 - - Use Firebase ML Kit for personalization
- 10.2 Vendor Marketplace Expansion
 - - 3rd party seller inventory management
- 10.3 Automation
 - - Firestore triggers for stock/order updates